Measuring Factor Crowding When Leverage Is Observable: Evidence from China's A-Share Market

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Abstract

Factor crowding is difficult to measure because holdings are rarely observed and return-based proxies may capture several forms of common movement. We investigate whether two proxies identify the same predictive object when leverage is observable. Using six style factors in China's A-share market from 2015–2025, we compare residual-return comovement with a margin-financing-balance proxy. Within-factor comovement between the two measures is limited. Comomentum is not robustly associated with subsequent factor returns; its one-month volatility diagnostic is directionally consistent but the broader second-moment evidence is exploratory. The margin-balance proxy is conditionally negatively associated with subsequent factor returns after controlling for factor momentum, balance levels, coverage, and factor fixed effects. In the main six-month specification, a one-standard-deviation increase is associated with approximately a four-percentage-point lower horizon-sum factor return. Because rolling formation windows overlap, we use pooled circular-shift permutation inference alongside HAC standard errors. The results support a measurement distinction: return comovement and observable leveraged financing need not be interchangeable measures of factor crowding, and the evidence is associational rather than causal.

Keywords: factor crowding, factor decay, comomentum, margin financing, China, A-share market, G12

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1. Introduction

A factor that attracts too much capital should, in the end, stop working. Arbitrage is limited, so capital flows into a strategy until the mispricing it exploits is gone, and the trades most exposed to a reversal are the crowded ones (Shleifer and Vishny, 1997; De Long et al., 1990; Stein, 2009; Barberis and Shleifer, 2003). What is not settled is the measurement: positions are unobserved, and the returns that would reveal them are contaminated by the price pressure crowding is supposed to create.

Lou and Polk (2022) propose a way around this. If many arbitrageurs hold the same stocks, those stocks should comove beyond what common factors explain. Their *comomentum* measure, the average pairwise correlation of three-factor residual returns within the stocks a momentum strategy trades, forecasts lower momentum returns, higher volatility and more negative skewness. It is indirect by construction: it infers positioning from second moments for want of anything better.

We ask whether that inference survives where an alternative is observable. China's A-share market combines a distinct margin-financing institution with tighter short-sale eligibility than the setting in which comomentum was developed. We therefore compare the return- correlation proxy with a balance-based proxy rather than treating either one as a definitive measure of crowding.

That comparison is the point. A correlation-based proxy and a balance-based proxy can produce different predictive results, making the interpretation of any single proxy difficult. We construct both for six style factors over 2015–2025 on the full A-share cross-section; the initial universe includes delisted firms. The two measures have limited linear comovement and give different answers about predictability: the margin-balance proxy is negatively associated with subsequent factor returns, whereas comomentum is not robustly predictive.

The first concerns a mechanical component of balance changes. A balance-based change measure, $\Delta MB_{i,t}/MV_{i,t}$, contains information about the balance level, which is related to market capitalisation and factor membership. We therefore control for the level throughout. The comparison between specifications A and C is a diagnostic for this mechanical component, not proof that all measurement error has been removed.

The second concerns factor momentum. Factors that performed well may attract financing, so crowding and past factor returns can move together.

We separate them by orthogonalising the proxy against the factor's past 1-, 6- and 12-month returns, including both variables simultaneously, and estimating factor-specific time-series regressions. Factor fixed effects ensure that the pooled coefficient is identified from within-factor variation rather than persistent differences in average crowding.

The third concerns inference. The formation window is twelve months but we sample it monthly, so adjacent observations share eleven twelfths of their data. We therefore report pooled circular-shift permutation p-values with the same estimator used for the observed statistic.

The paper contributes to two literatures. To the literature on factor crowding and decay (McLean and Pontiff, 2016; Beck et al., 2016; Lou and Polk, 2022), it adds a measurement comparison under a different arbitrage regime. The return-correlation and financing-balance proxies span six factors and are evaluated within the same market. To the literature on China's margin trading (Bian et al., 2026; Xing et al., 2026), it adds a factor-level association between financing balances and subsequent factor returns. The contribution is the measurement comparison and its inferential design, not a claim that either proxy directly observes total investor positions.

2. Data and methodology

2.1. Data

We use the full A-share cross-section from January 2015 to December 2025. Daily prices, trading volumes, turnover, and point-in-time valuation ratios are from Baostock; the universe includes delisted firms. The sample covers 5,247 stocks over 2,674 trading days. Individual-stock margin-financing balances, covering the designated margin-eligible list, are from the Shanghai and Shenzhen exchanges.

We treat the raw margin date as an occurrence date and impose a conservative next-trading-day availability rule. The common month-end signal is formed only from balances available by that date; the resulting audit has no observations available after the signal date. The valuation ratios are treated as point-in-time fields based on the multi-stock audit documented in the replication materials.

We exclude stocks that are suspended, have been listed fewer than 180 days, or are flagged ST, and we drop financial firms using the available industry classification. Within the monthly cross-section we additionally drop stock-months with more than ten non-trading days, and we apply the same

rule when forming the decile legs that define the crowding measures. The margin-eligible list expands materially over the sample. We therefore construct factor portfolios on the full cross-section rather than within the eligible pool, and compute the margin-balance proxy only over constituent stocks, with coverage controls for the changing eligible universe.

2.2. Two measures of crowding

Indirect: comomentum.. Following Lou and Polk (2022), for each factor f and month t we take the long and short deciles formed on f's exposure at t, regress each constituent's daily returns over the trailing 250 trading days on a three-factor benchmark constructed from the cleaned universe, and compute the average pairwise correlation of the residual returns within each leg. The measure is the average across the two legs. It is a return-correlation proxy, not an account-level holdings measure. Because each leg is capped at 100 randomly selected stocks when necessary, it is also a sampled pairwise-correlation measure; the sampling seed is fixed for replication.

Margin-financing-balance proxy.. For each constituent we compute the monthly change in the publicly reported margin-financing– balance scaled by float market capitalisation, $\Delta MB_{i,t}/MV_{i,t}$, and aggregate to the leg level using float-capitalisation weights. The measure is the long-leg aggregate minus the short-leg aggregate. We also retain the long leg separately, the aggregate margin level, and the portfolio's margin coverage. The raw balance date is treated as an occurrence date; under the exchange disclosure rules, the previous trading day's information is published before the next trading session, so we use the next trading day as a conservative availability date. This proxy is more directly observed than comomentum, but it is not a pure financing flow, an account-level position, or a causal measure of crowding.

2.3. Empirical design

For each measure $C_{f,t}$ and horizon $h \in \{1, 3, 6, 12\}$ we estimate

$$R^{f}_{t\to t+h} \;=\; \alpha_f \;+\; \beta\, C_{f,t} \;+\; \gamma' X_{f,t} \;+\; \varepsilon_{f,t}, \tag{1}$$

where $R^{f}_{t\to t+h}$ is the horizon sum of the next h monthly long-short returns, starting in $t+1$ (we report compound-return sensitivity separately), α_f is a factor fixed effect, and X collects the factor's own past 1-, 6-, and 12-month returns, the margin level, and the portfolio's margin coverage. Standard

errors are Newey-West with six lags. Because the six factors share a market shock within a month, factor-level clustering is not reliable with so few clusters, and we do not rely on it.

Our central concern is that β reflects factor momentum rather than crowding. We address it three ways: by orthogonalising $C_{f,t}$ against the factor's own past 1-, 6-, and 12-month returns and re-running the regression on the residual; by including crowding and past returns simultaneously; and by estimating the regression separately for each factor, which asks whether the sign is common or driven by one factor.

Finally, because the formation window is twelve months but sampling is monthly, adjacent observations share eleven twelfths of their data. For each permutation we shift the measure within factor and re-estimate the same pooled specification, preserving factor fixed effects, controls, the sample, and the HAC procedure. We use the finite-permutation correction $(1 + B_{\geq})/(1 + B)$ for the reported p-values and treat them as the primary inferential quantities; HAC p-values are shown for comparison.

We validate the correction rather than assert it. On synthetic panels with no crowding effect by construction, across three independent draws the revised pooled permutation procedure provides a calibration check; the complete calibration output and seeds are in the replication materials. Since a null result is only informative if the design has power, we also compute minimum detectable effects by block bootstrap, injecting known effects into the actual panel and recording the rejection rate at the permutation critical value.

The primary evidence family is the two proxies crossed with Layers 1 and 3 and horizons $h \in \{1, 3, 6\}$. Horizon 12, factor-specific regressions, tail analysis, second-moment diagnostics, and robustness splits are auxiliary. We use finite pooled circular-shift permutation p-values for the primary family and report HAC statistics for comparison.

3. Results

3.1. The two measures of crowding are only weakly related

The two measures have limited linear comovement, although this alone does not establish that they identify distinct economic objects. Table 1 reports the within-factor correlation between comomentum and the margin-balance proxy. It ranges from -0.29 for turnover to $+0.27$ for size, and is near zero for low-volatility. Pooling the six factors and removing factor-specific

Table 1: Comomentum and margin crowding: correlation by factor

Factor	N	Pearson	Spearman	Mean comomentum	Mean margin-balance proxy
VAL	120	-0.269	-0.159	0.02307	-0.0008
SMB	120	0.266	0.249	0.00638	-0.0001
MOM	120	-0.168	-0.151	0.02656	0.0010
QMJ	120	0.052	-0.013	0.01439	0.0002
LOWVOL	120	0.008	0.009	0.02970	-0.0019
TURN	120	-0.252	-0.192	0.02043	-0.0037
Pooled	720	-0.141	-0.144	0.02009	-0.0009
Pooled (factor-demeaned)	720	-0.093	-0.073	0.00000	0.0000

means leaves -0.11. These correlations are descriptive and may also reflect timing, coverage, nonlinear dependence, or sampling noise.

3.2. Margin-balance proxy predicts decay; comomentum does not

The primary coefficients and finite permutation p-values are reported in Table 2; HAC t-statistics are retained in the replication outputs. Comomentum has no robust predictive association with subsequent factor returns in this sample and design: permutation p-values run from 0.30 to 0.95, and the point estimates change sign across horizons. The margin-balance proxy, by contrast, is negatively associated with subsequent returns, with the effect remaining after controls are added.

The pattern across specifications is informative. With no controls (column A) the margin-balance coefficient is negative but imprecisely estimated. Adding the factor's own past 1-, 6- and 12-month returns (column B) increases the coefficient's absolute magnitude; at $h = 6$, β is about -0.034 and the permutation p-value is about 0.010. Controlling for the margin level (C) and coverage (D) leaves the estimates essentially unchanged. This comparison diagnoses, but does not eliminate, concerns about the level component of balance changes.

The magnitude is economically material in this descriptive sense: in specification D at $h = 6$, a one-standard-deviation increase in the margin-balance proxy accompanies approximately a 4.0 percentage point lower horizon-sum long-short return over six months. This is a conditional predictive association, not an estimate of a causal effect.

3.3. The return-correlation proxy is not merely a failed implementation

The obvious objection to the previous section is that our implementation of comomentum fails, rather than the proxy itself. Lou and Polk (2022) also

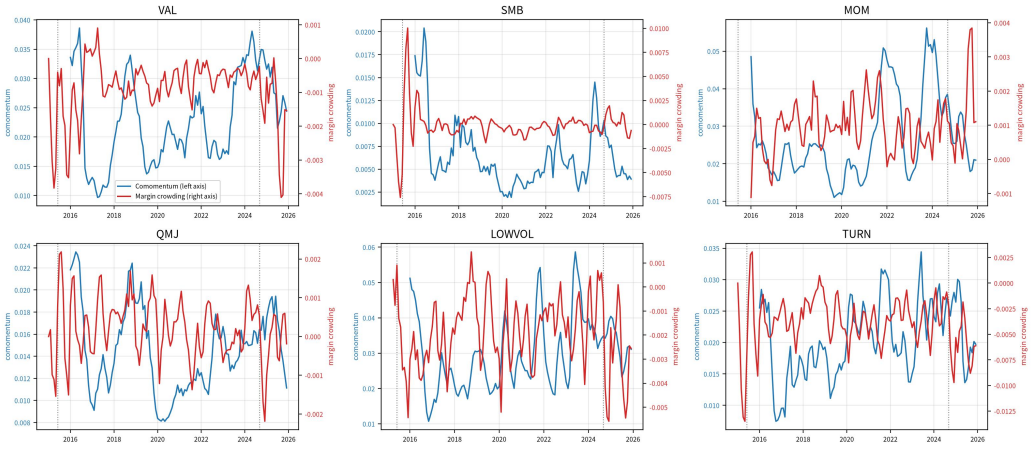


Figure 1: The two crowding measures by factor, monthly, three-month rolling mean. Comomentum (blue, left axis) is the average pairwise correlation of three-factor residual returns within the factor's long and short legs. Margin crowding (red, right axis) is the float-weighted change in margin financing balances on the long leg less the short leg, scaled by float market capitalisation. Dotted vertical lines mark June 2015 and September 2024. The two series are plotted on separate axes because their units differ by one to two orders of magnitude.

document that comomentum forecasts higher realised volatility and more negative skewness. Table 3 shows some directional consistency in China: the one-month volatility coefficient has permutation $p = 0.059$, while the skewness results are not individually significant under the revised pooled permutation procedure. This is supportive diagnostic evidence, not a full replication or proof that the proxy identifies account-level crowding.

The margin-balance proxy shows a different pattern: it is negatively associated with returns and has no reliable relation to the second moments in the primary specifications. The two proxies therefore need not be interchangeable. Comomentum tracks residual-return comovement, whereas the margin-balance proxy tracks a balance-based financing signal.

3.4. Separating crowding from factor momentum

Table 4 reports the defenses against the interpretation that β is factor momentum. Orthogonalising the proxy against past returns and the margin level and re-running the regression leaves the primary margin-balance association intact. Including the proxy and past returns simultaneously gives the same qualitative result. Factor-specific slopes are reported as descriptive

evidence only; with six factors, they do not provide a reliable stand-alone inferential test.

3.5. The effect is in the centre, not the tail

A natural competitor to the mean-shift story is that the margin-balance proxy raises crash risk; in this sample the auxiliary evidence does not establish that pattern. Table 6 reports pooled quantile regressions with factor fixed effects, and Table 5 compares the two crowding groups' distributions. The largest negative coefficients occur near the median and upper quantiles, while the left-tail coefficient is not significant under the revised procedure. These results are auxiliary and do not constitute a formal test of quantile-coefficient equality or a point-in-time crash threshold.

3.6. Robustness and scope of the null result

Table 7 splits the sample, changes the weighting, and varies the construction window. The margin-balance association is weak in 2015–2018 ($p = 0.33$), not retained at the 0.10 level in 2019–2021 ($p = 0.11$) or 2022–2025 ($p = 0.15$), and remains detectable under equal weighting ($p = 0.008$). It weakens when the balance change is measured over six or twenty-four months and for the long leg alone. These results show directional and design sensitivity; they should not be described as universal robustness.

Because one of the two results is a null, its informativeness matters. Under the revised pooled permutation design, minimum detectable effects from the block bootstrap are about 11–14 per cent annualised for comomentum across the reported horizons. The null is informative relative to effects above this design-specific detection threshold; it does not establish equivalence to zero.

4. Conclusion

Two proxies for factor crowding, applied to the same factors in the same market, give substantially different predictive results. The return-correlation proxy has no robust association with subsequent factor returns in this sample and design. The margin-balance proxy is negatively associated with subsequent returns after controls for factor momentum, the balance level, coverage, factor fixed effects, and pooled circular-shift permutation inference. The result is conditional and associational, not a causal estimate or an account-level holdings result.

The evidence supports a measurement distinction. Comomentum provides a residual-return comovement signal with limited directional evidence in the second-moment diagnostics, while the margin-balance proxy provides the stronger return-predictive association. The two proxies should therefore be reported side by side rather than treated as interchangeable. The subsample and window results show that the association is not universal, and the auxiliary tail analysis does not establish disproportionate crash risk.

Trailing-window measures sampled monthly create overlapping observations. The pooled circular-shift procedure preserves each factor's measure persistence and re-estimates the same specification for every shift; it is the primary inferential procedure here. Future work should use historical industry classifications, account- or position-level data, and quasi-experimental variation in margin eligibility or disclosure.

Table 2: Crowding and future factor returns

Coefficient (stars from permutation p)					
Measure	h	A	B	C	D
Comomentum	1	0.0019	0.0021	0.0031	0.0036
Comomentum	3	0.0011	0.0037	0.0068	0.0082
Comomentum	6	0.0014	0.0059	0.0113	0.0131
Comomentum	12	-0.0211	-0.0183	-0.0107	-0.0102
Margin-balance proxy	1	-0.0087***	-0.0137***	-0.0150***	-0.0157***
Margin-balance proxy	3	-0.0115	-0.0226***	-0.0264***	-0.0284***
Margin-balance proxy	6	-0.0177	-0.0338***	-0.0383***	-0.0400***
Margin-balance proxy	12	-0.0190	-0.0377*	-0.0469**	-0.0479**

Notes: Stars use finite circular-shift permutation p-values; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3: Crowding and the second moments of factor returns

Panel A: coefficient			
Measure	h	Volatility	Skewness
Comomentum	1	0.0014*	-0.0909
Comomentum	3	0.0009	-0.1312
Comomentum	6	0.0006	-0.1242
Margin-balance proxy	1	-0.0002	-0.0236
Margin-balance proxy	3	-0.0006*	-0.0390
Margin-balance proxy	6	-0.0005	-0.0326
Panel B: permutation p-value			
Measure	h	Volatility	Skewness
Comomentum	1	0.059	0.288
Comomentum	3	0.216	0.310
Comomentum	6	0.407	0.345
Margin-balance proxy	1	0.737	0.602
Margin-balance proxy	3	0.095	0.302
Margin-balance proxy	6	0.115	0.239

Notes: Stars use finite circular-shift permutation p-values; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Separating crowding from factor momentum

Measure	Layer	h	β	HAC t	Perm. p
Comomentum	Layer 1: orthogonalised residual	1	0.0031	0.87	0.374
Comomentum	Layer 1: orthogonalised residual	3	0.0068	0.72	0.610
Comomentum	Layer 1: orthogonalised residual	6	0.0117	0.72	0.706
Comomentum	Layer 1: orthogonalised residual	12	-0.0077	-0.27	0.906
Comomentum	Layer 3: all controls simultaneously	1	0.0031	0.87	0.355
Comomentum	Layer 3: all controls simultaneously	3	0.0068	0.76	0.590
Comomentum	Layer 3: all controls simultaneously	6	0.0113	0.75	0.539
Comomentum	Layer 3: all controls simultaneously	12	-0.0107	-0.43	0.781
Margin-balance proxy	Layer 1: orthogonalised residual	1	-0.0150***	-3.95	0.009
Margin-balance proxy	Layer 1: orthogonalised residual	3	-0.0260***	-3.69	0.010
Margin-balance proxy	Layer 1: orthogonalised residual	6	-0.0392***	-3.76	0.010
Margin-balance proxy	Layer 1: orthogonalised residual	12	-0.0469*	-2.69	0.052
Margin-balance proxy	Layer 3: all controls simultaneously	1	-0.0150***	-4.09	0.009
Margin-balance proxy	Layer 3: all controls simultaneously	3	-0.0264***	-4.10	0.010
Margin-balance proxy	Layer 3: all controls simultaneously	6	-0.0383***	-3.78	0.010
Margin-balance proxy	Layer 3: all controls simultaneously	12	-0.0469**	-2.98	0.010
Comomentum	Layer 2: within-factor time series	1	0.0018	0.32	0.776
Comomentum	Layer 2: within-factor time series	1	-0.0062	-0.69	0.551
Comomentum	Layer 2: within-factor time series	1	-0.0215	-1.98	0.118
Comomentum	Layer 2: within-factor time series	1	0.0063	0.50	0.723
Comomentum	Layer 2: within-factor time series	1	0.0108	1.10	0.345
Comomentum	Layer 2: within-factor time series	1	0.0142	2.08	0.109
Comomentum	Layer 2: cross-factor mean slope	1	0.0009	0.17	—
Margin-balance proxy	Layer 2: within-factor time series	1	-0.0130	-1.41	0.181
Margin-balance proxy	Layer 2: within-factor time series	1	-0.0406**	-3.78	0.019
Margin-balance proxy	Layer 2: within-factor time series	1	-0.0042	-0.47	0.672
Margin-balance proxy	Layer 2: within-factor time series	1	0.0026	0.25	0.874
Margin-balance proxy	Layer 2: within-factor time series	1	-0.0193**	-2.98	0.042
Margin-balance proxy	Layer 2: within-factor time series	1	-0.0186	-1.05	0.328
Margin-balance proxy	Layer 2: cross-factor mean slope	1	-0.0155	-2.54	—

Notes: Stars use finite circular-shift permutation p-values; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: Crowding and the distribution of factor returns

Measure	Factor	Crowding	Mean	SD	Skew.	Kurt.	Crash freq.
Comomentum	VAL	High	0.0127	0.0582	0.38	0.83	0.02
	VAL	Low	-0.0089	0.0748	-0.40	0.16	0.07
	SMB	High	0.0091	0.0635	0.13	0.04	0.03
	SMB	Low	-0.0021	0.0651	-0.36	0.90	0.08
	MOM	High	0.0007	0.0684	-0.43	0.42	0.07
	MOM	Low	0.0143	0.0734	-0.08	0.91	0.03
	QMJ	High	-0.0057	0.0514	-1.23	2.68	0.07
	QMJ	Low	0.0134	0.0545	-0.37	0.67	0.03
	LOWVOL	High	0.0087	0.0907	-0.07	1.00	0.07
	LOWVOL	Low	0.0113	0.0685	-0.29	0.16	0.03
	TURN	High	0.0172	0.0960	-0.83	2.59	0.05
	TURN	Low	0.0225	0.0660	0.26	0.81	0.02
Margin-balance proxy	VAL	High	0.0002	0.0696	-0.13	0.67	0.03
	VAL	Low	0.0013	0.0832	-0.54	2.51	0.08
	SMB	High	0.0071	0.0663	0.25	0.32	0.05
	SMB	Low	0.0136	0.0895	1.59	6.44	0.06
	MOM	High	-0.0028	0.0684	-0.30	0.40	0.07
	MOM	Low	0.0179	0.0725	-0.20	1.06	0.03
	QMJ	High	0.0023	0.0534	-0.30	-0.03	0.05
	QMJ	Low	0.0029	0.0544	-0.90	2.71	0.06
	LOWVOL	High	0.0112	0.0835	-0.06	1.35	0.06
	LOWVOL	Low	0.0094	0.0825	-0.26	0.59	0.05
	TURN	High	0.0160	0.0715	-0.58	2.05	0.03
	TURN	Low	0.0191	0.0993	-0.65	1.60	0.08

Table 6: Quantile regressions: does crowding move the tail more than the centre?

Measure	$q = 0.10$	$q = 0.25$	$q = 0.50$	$q = 0.75$	$q = 0.90$
Comomentum	0.0010	-0.0006	-0.0037	0.0037	0.0098
Margin-balance proxy	0.0011	-0.0020	-0.0093**	-0.0097*	-0.0107
Permutation p-values					
Comomentum	0.798	0.866	0.361	0.429	0.151
Margin-balance proxy	0.916	0.710	0.031	0.061	0.145

Table 7: Robustness

Check	Grouping	Measure	β	HAC t	Perm. p	N
Subsample	2015–2018 leverage cycle	Comomentum	-0.0026	-0.51	0.583	216
Subsample	2015–2018 leverage cycle	Margin-balance proxy	-0.0046	-1.25	0.194	273
Subsample	2019–2021 foreign inflows	Comomentum	0.0043	0.46	0.750	216
Subsample	2019–2021 foreign inflows	Margin-balance proxy	-0.0241	-3.04	0.111	216
Subsample	2022–2025 retail margin	Comomentum	0.0101	1.72	0.128	282
Subsample	2022–2025 retail margin	Margin-balance proxy	-0.0103	-2.32	0.149	282
Equal-weighted	Equal-weighted	Comomentum	0.0000	0.01	0.992	714
Equal-weighted	Equal-weighted	Margin-balance proxy	-0.0072***	-3.01	0.008	771
Window	Balance change, 6 months	Margin-balance proxy	-0.0076	-1.33	0.185	771
Window	Balance change, 24 months	Margin-balance proxy	-0.0053	-1.50	0.244	771
Variant	Long leg only	Margin-balance proxy, long leg	-0.0011	-0.61	0.672	771

Notes: Stars use finite circular-shift permutation p-values; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Author declarations

Data availability

The daily market and valuation data are obtained from Baostock, and the margin-financing data are obtained from public exchange disclosures. Redistribution of raw provider data may be subject to upstream terms. The analysis code, configuration, evidence manifest, run manifest, derived-data documentation, and derived summary outputs will be made available with the replication package.

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Declaration of competing interests

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Declaration of generative AI use

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References

Barberis, N., Shleifer, A., 2003. Style investing. Journal of Financial Economics 68, 161–199. doi:10.1016/S0304-405X(03)00064-3.

Beck, N., Hsu, J., Kalesnik, V., Kostka, H., 2016. Will your factor deliver? an examination of factor robustness and implementation costs. Financial Analysts Journal 72, 63–74. doi:10.2469/faj.v72.n5.6.

Bian, J., Da, Z., He, Z., Lou, D., Shue, K., Zhou, H., 2026. The drivers and implications of retail margin trading. The Journal of Finance 81, 2217–2270. doi:10.1111/jofi.70049.

De Long, J.B., Shleifer, A., Summers, L.H., Waldmann, R.J., 1990. Noise trader risk in financial markets. Journal of Political Economy 98, 703–738. doi:10.1086/261898.

Lou, D., Polk, C., 2022. Comomentum: Inferring arbitrage activity from return correlations. The Review of Financial Studies 35, 3272–3302. doi:10.1093/rfs/hhab117.

McLean, R.D., Pontiff, J., 2016. Does academic research destroy stock return predictability? The Journal of Finance 71, 5–32. doi:10.1111/jofi.12365.

Shleifer, A., Vishny, R.W., 1997. The limits of arbitrage. The Journal of Finance 52, 35–55. doi:10.1111/0022-1082.00001.

Stein, J.C., 2009. Presidential address: Sophisticated investors and market efficiency. The Journal of Finance 64, 1517–1548. doi:10.1111/j.1540-6261.2009.01472.x.

Xing, S., Luo, Y., Li, T., 2026. PMO factor crowding and returns: Evidence from china. Applied Economics Letters 33, 987–993. doi:10.1080/13504851.2024.2401517.